

Contributors

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Scheduling Algorithms

Round Robin (RR) scheduling algorithm

- Each process in the ready queue gets a small unit of CPU time (time quantum (time slice): q), usually 10-100 milliseconds.
- After this time has elapsed, the process is preempted and added to the end of the ready queue.
- New processes are added to the tail of the ready queue.

One of two things happens during process execution:

1 If CPU burst of the current running process is less than 1 time quantum:

- In this case, the process itself will release the CPU voluntarily (freely).
- The CPU scheduler will then proceed to the next process in the ready queue.

Scheduling Algorithms

2- If the CPU burst of the currently running process is longer than 1 time quantum,

- The timer will go off and will cause an interrupt to the operating system.
- The running process will be put at the tail of the ready queue.
- The CPU scheduler will then select the next process in the ready queue

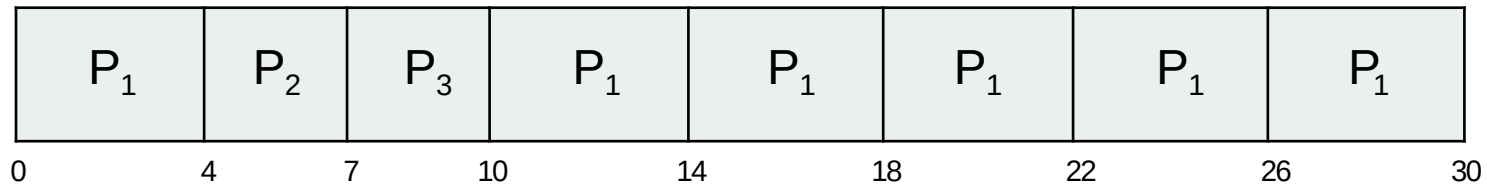
Scheduling Algorithms

Round Robin scheduling algorithm is preemptive:

If a process's CPU burst exceeds 1 time quantum, that process is preempted and is put back in the ready queue.

Scheduling Algorithms

Ex.: If Time Quantum = 4 milliseconds



Process	Burst Time
P1	24
P2	3
P3	3

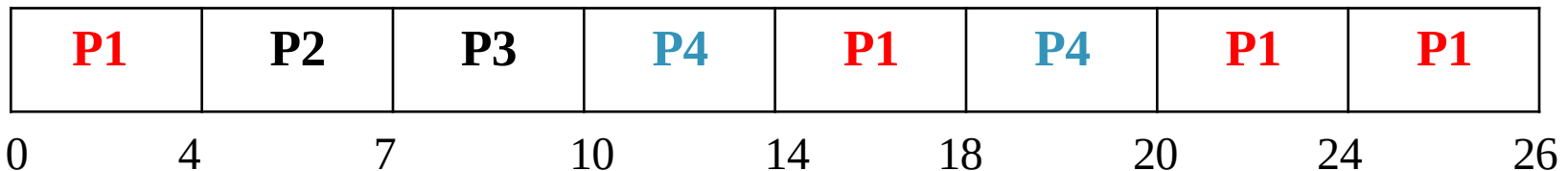
Only occurs in Round Robin to have more than one waiting time

- Waiting time for P1 = $10 - 4 = 6$
- Means when it is started for the second time (at 10) minus its last finish (at 4)
- Turnaround time for P1 = $6 + 24 = 30$

Scheduling Algorithms

Another example in Round Robin when time quantum =4

Process	Burst Time
P1	14
P2	3
P3	3
P4	6



- Waiting time for p1 = $0 + (14 - 4) + (20 - 18) = 0 + 10 + 2 = 12$ ms
- Turnaround time for P1 = waiting + burst = $12 + 14 = 26$
- Waiting time for P4 = $10 + (18 - 14) = 10 + 4 = 14$
- Turnaround time for P4 = $14 + 6 = 20$